

Groups, Fields, and Vector Spaces

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Definition of a Group

A **group** is a non-empty set G together with a binary operation

$$* : G \times G \rightarrow G$$

such that for all $a, b, c \in G$, the following axioms hold:

- ① **Closure:** $a * b \in G$
- ② **Associativity:** $(a * b) * c = a * (b * c)$
- ③ **Identity:** $\exists e \in G$ such that $a * e = e * a = a$
- ④ **Inverse:** $\forall a \in G, \exists a^{-1} \in G$ such that $a * a^{-1} = e$

Examples of Groups

- $(\mathbb{Z}, +)$: Integers under addition
- $(\mathbb{R}, +)$: Real numbers under addition
- $(\mathbb{R} \setminus \{0\}, \cdot)$: Nonzero real numbers under multiplication
- $(\mathbb{Z}_n, +_n)$: Integers modulo n

Note: $(\mathbb{Z}, \cdot)$ is **not** a group since inverses do not exist for all elements.

Definition of a Field

A **field** is a set F together with two operations $+$ and $\cdot$ such that:

- $(F, +)$ is an abelian group with identity 0
- $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1
- Multiplication distributes over addition:

$$a(b + c) = ab + ac, \quad \forall a, b, c \in F$$

Examples of Fields

- $\mathbb{Q}$: Rational numbers
- $\mathbb{R}$: Real numbers
- $\mathbb{C}$: Complex numbers
- $\mathbb{Z}_p$: Integers modulo a prime p

Remark: $\mathbb{Z}_n$ is a field **if and only if** n is prime.

Definition of a Vector Space

Let V be a non-empty set and F be a field. The set V is called a **vector space over F** if two operations are defined:

- Vector addition: $+: V \times V \rightarrow V$
- Scalar multiplication: $\cdot: F \times V \rightarrow V$

such that for all $u, v, w \in V$ and $a, b \in F$, the following axioms hold.

Vector Space Axioms

Axioms of Vector Addition

- ① $u + v = v + u$
- ② $(u + v) + w = u + (v + w)$
- ③ $\exists 0 \in V$ such that $u + 0 = u$
- ④ $\forall u \in V, \exists (-u) \in V$ such that $u + (-u) = 0$

Axioms of Scalar Multiplication

- ① $a(u + v) = au + av$
- ② $(a + b)u = au + bu$
- ③ $(ab)u = a(bu)$
- ④ $1u = u$

Examples of Vector Spaces

- $\mathbb{R}^n$ over $\mathbb{R}$
- $\mathbb{C}^n$ over $\mathbb{C}$
- Set of all polynomials P_n over $\mathbb{R}$
- Set of all $m \times n$ matrices over $\mathbb{R}$
- Set of all continuous functions on $[a, b]$

Definition of a Subspace

Let V be a vector space over a field F . A non-empty subset $W \subseteq V$ is called a **subspace of V** if W itself is a vector space under the same operations of V .

Subspace Test

A non-empty subset $W \subseteq V$ is a subspace of V if and only if:

- 1 **Closure under addition:** If $u, v \in W$, then $u + v \in W$
- 2 **Closure under scalar multiplication:** If $u \in W$ and $a \in F$, then $au \in W$

Remark: The zero vector $0 \in V$ must belong to W .

Examples of Subspaces

- $\{0\}$ (trivial subspace)
- $\mathbb{R}^n$ itself
- Set of all solutions of a homogeneous linear system
- Set of all polynomials with zero constant term
- Set of all diagonal matrices in $M_n(\mathbb{R})$

Non-Examples of Subspaces

- Set of vectors in $\mathbb{R}^2$ with first coordinate 1
- Set of all matrices with determinant 1
- Set of all polynomials of degree exactly 2

Reason: These sets are not closed under addition or scalar multiplication.

Conclusion

- Vector spaces generalize algebraic and geometric structures
- Subspaces preserve vector space structure
- Subspace test simplifies verification significantly

Conclusion

- Groups capture the idea of symmetry and structure
- Fields allow both addition and multiplication
- Vector spaces generalize geometric and algebraic objects
- Subspaces preserve vector space structure
- Subspace test simplifies verification significantly

These structures form the foundation of modern algebra and applied mathematics.

Thank You